

UI/UX Design

Document 03

Status	Draft v1
Version	1.1 · 2026-10-02 00:34 · change log
Stack	FastAPI backend serving a plain HTML/CSS/JavaScript page (no framework); built first for the Transcribe surface (python -m transcribe.app)
Related	01 PRD , 04 App Flow

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1 Design principles

- 1. **Verify, don't type.** Every text box opens pre-filled (aligned transcript or model hypothesis). The fastest action is “approve”.
- 2. **Keyboard first.** A reviewer should process a segment without touching the mouse.
- 3. **The model suggests, the human decides.** Model output is always visually distinct (a diff panel) and is never silently written into the editable text.
- 4. **One primary action per screen,** shown with the primary colour. Destructive actions are secondary and need confirmation.
- 5. **State is always visible.** The segment's status, whether there are unsaved changes, job progress and the active model are shown on screen.

2 Visual language

This keeps the earth-tone theme already chosen in review/app.py and formalises it as tokens.

Token	Light	Dark	Use
--bg	#f4eee5	#211b17	page
--surface	#fffaf3	#30261f	cards, inputs
--text	#2d241d	#f4eee5	body

Token	Light	Dark	Use
--muted	#675443	#d5bda5	secondary text, timestamps
--border	#b99b7a	#806149	dividers
--accent	#8a5a3b	#b77b51	primary buttons, focus ring
--accent-hover	#6f432d	#d19669	
--ok	#4f7a3a	#8fbf6f	approved
--warn	#b07d1a	#e0b04f	flagged, low confidence
--danger	#9c3b2e	#e07a6a	rejected, errors
--lang-en	#2f5d8a underline	#7fb0e0	language span tint
--lang-sw	#4f7a3a underline	#8fbf6f	
--lang-mixed	#8a3b7a underline	#d08ac0	

Typography: Times New Roman (the project's choice) for UI text at 12 pt minimum. The transcript editor uses **14 pt** with a line height of 1.6, because reviewers read it for hours. Timestamps and IDs use a monospace font at 11 pt. Colour is never the only signal: flags also show an icon and a label.

3 Information architecture

```

Top bar: [Transcribe] [Corpus Studio] [Models & Evaluation]      active model: whisper-small-
↳ kct-v1 ●
  |
  |-- Transcribe
  |   |-- New transcription (upload)
  |   |-- Jobs list
  |   `-- Transcript editor
  |-- Corpus Studio
  |   |-- Recordings list
  |   |-- Import recording (wizard)
  |   |-- Recording overview (alignment report)
  |   |-- Review (segment reviewer)  <- most-used screen
  |   `-- Datasets (build / export snapshots)
  `-- Models & Evaluation
      |-- Model registry
      |-- Evaluation report
      `-- Compare two models

```

4 Screens

4.1 Transcribe: New transcription

Changed in v1.1: see the [change log](#).

Two pages: **Load audio**, then **Transcription in progress**.

```

+-----+
| Transcription                                     |
| +-----+                                         |
| |                               Load audio       | |
| | Drop a file here or click to choose. Any audio or video. | |
| +-----+                                         |
| KSM09.mp3  31.2 MB      [ > preview player ..... ] |
| Language [Kiswahili + English v]  Model [Whisper small v] |
| [x] Label speakers when possible                   |

```

```

| [                               Transcribe                               ] |
+-----+
| Recent transcriptions: KSM09 · 12 of 240 chunks corrected · ... |
+-----+

+-----+
|                               ( spinner )                               |
|                               Transcription in progress                 |
| Please be patient. This runs in the background: you can leave         |
| this page and come back later from Recent transcriptions.             |
|                               Transcribing: chunk 37 of 240              |
|                               [#####.....]                          |
|                               About 14:20 left                          |
+-----+

```

- Any audio or video format is accepted (decoded with PyAV). The selected file can be previewed before uploading, and the upload shows its own progress bar.
- The language setting defaults to Kiswahili + English, the setting that worked best on code-switched speech in the pilot; English and automatic detection are alternatives. The model is Whisper small (faster) or large-v3-turbo (more accurate, slower).
- Speaker labels (Speaker 1, Speaker 2, ...) appear when pyannote is installed and a Hugging Face token is set; otherwise chunks have no speaker.
- The progress page shows the stage (preparing audio, finding speech, transcribing chunk n of N , identifying speakers) and an estimate of the time left, then opens the editor by itself.

4.2 Transcribe: Transcript editor

Changed in v1.1: [see the change log](#).

Chunk by chunk: the audio is split at pauses into chunks of up to 30 s, and each chunk is played and corrected on its own.

```

+-----+
| Transcription  KSM09                               12 of 240 corrected  Export |
+-----+
| ✓ 11 2:41 Leo tu.. | Chunk 13 of 240  3:05 - 3:19  Speaker 2  | | |
| > · 13 3:05 Na sa.. | [re-transcribed with your corrections] |
| ♪ 14 3:20 Kwa sa.. | (||) ●===== 0:06 / 0:14 |
| ♪ 15 3:34 Halafu.. |                               1.10x |
| · 16 3:51 ...      | +-----+ |
|                    | | Na sasa wale wa nyuma mhasikia vizuri? | |
|                    | +-----+ |
|                    | > Model text |
|                    | [<- Previous]   Saved   [Save & next ->]|
|                    | Learned: sababa -> sababu (applied to 14 |
|                    | chunks). Re-transcribing the next 3 chunks.|
|                    | Esc play/pause  F1 back 3 s  F2 forward 3 s|
|                    | F3 slower  F4 faster  Ctrl+Enter save&next |
+-----+

```

- **Keys** (shared with the Corpus Studio reviewer, §4.5): Esc play/pause; F1/F2 back/forward 3 s; F3/F4 slower/faster ($0.5\times$ – $2\times$, pitch preserved, kept across chunks); Ctrl+Enter save and next; Alt+←/Alt+→ previous/next; Ctrl+S save.
- **No play button needed:** moving to another chunk saves the current edit and starts playing the new chunk. The next chunk's audio is preloaded.
- **Chunk list:** ✓ corrected, ♪ improved by your corrections, · model text only. Clicking a chunk opens it.

- **Learning while you work** (TRD §5): a word corrected the same way twice becomes a rule applied to every chunk not yet edited; your recent corrected text and new words are given to Whisper as context, and the next 3 chunks are re-transcribed in the background. Improved text replaces a chunk's text only if you have not typed in it; a corrected chunk is never changed.
- **Model text** (collapsed) shows Whisper's own output for comparison. Export: TXT (grouped by speaker), SRT, JSON.

4.3 Corpus Studio: Import recording (3-step wizard)

1. **Files:** audio (required), transcript DOCX/PDF (optional).
2. **Metadata:** title, domain, recorded date, region (free text, optional), consent scope (required: *training / eval only / transcription only*), and the language-hint convention for the DOCX ("italic = Kiswahili" ✓).
3. **Confirm:** a parsed preview showing the first 10 turns with speaker labels, the count of turns, the count of [unclear] markers, and the audio duration. Pressing **Import & align** enqueues the job.

4.4 Corpus Studio: Recording overview

KSM09 · interview · 1:12:04 · consent: training
 Alignment: done · mean confidence 0.81 · 612 segments 
 [] 402 approved · 31 flagged · 179 to review
 Low-confidence turns (likely non-verbatim): p0142 0.31 | p0388 0.28 | ...
 Confidence timeline:  (click to jump)
 [Start reviewing ►]

The confidence timeline shows problem regions (non-verbatim passages, music, crosstalk) at a glance.

4.5 Corpus Studio: Segment reviewer (primary screen)

```
+-----+
| KSM09 · segment 143/612 · priority 0.71 · status: candidate |
| Queue: [Priority v]  Filter: [All flags v]      unsaved ●    |
+-----+
| waveform  00:14:02.35 → 00:14:13.90 |
| ◀◀[ -250ms] ◀[ -50ms] start [+50ms ▶]  end ◀[ -50ms] [+50ms ▶]|
| ▶ Play (Tab)  ⌂ Replay last 3s (Shift+Tab)  0.75x / 1x    |
+-----+
| Transcript (editable)                                     14 pt |
| | Sasa mimi nilikuwa nimeapply hiyo job lakini they never |
| | called back, so nikaamua tu kuanza biashara yangu.      |
| | _____                                              |
| Source: aligned transcript -p0141p0142 · word conf 0.42 "nimeapply"|
+-----+
| Model hypothesis (whisper-small-kct-v1)      [Run model (R)] |
| Sasa mimi nilikuwa nimeapply hiyo job lakini they never      |
| called back so ~nikaamua~ **nikamua** tu kuanza biashara yangu|
| WER vs transcript: 7.1%  [Use hypothesis text (Ctrl+H)]      |
+-----+
| Flags: [U]nclear [O]verlap [N]oisy [M]usic [V]non-verbatim    |
| Language spans (select text, then E / S / X mixed / L Luo)   |
| Speaker: interviewer ▼                                       |
+-----+
| [Reject (Ctrl+Del)]  [Save draft (Ctrl+S)]  [Approve & next ↵] |
| History ▼  r3 you 14:02 · r2 aligner · r1 aligner             |
+-----+
```

Behaviour:

- **Approve & next** (Ctrl+Enter) saves a revision with status=approved and loads the next item in the queue. Tab plays the segment. The segment auto-plays on load (a setting, on by default).
- Changing a boundary re-renders the slice and replays the last 3 s (for the end boundary) or the first 3 s (for the start boundary).
- **Use hypothesis text** is the only way hypothesis text enters the editor. It is explicit, and it creates a revision with source=model_hypothesis.
- Flags are toggles, with the key shown in brackets. Setting *Unclear* or *Non-verbatim* shows an inline note: “Excluded from training until resolved.”
- Pressing Approve on a segment that has an active exclusion flag shows a confirmation: “Approve for eval only?”
- The History drawer lists revisions and offers “Restore r2” (which creates r4 as a copy of r2, so nothing is ever deleted).
- A 409 conflict shows a banner: “This segment changed in another tab. Reload / Keep mine as new revision.”

Keyboard map:

Key	Action
Esc	Play / pause (same as the transcript editor, §4.2)
F1 / F2	Back / forward 3 s
F3 / F4	Slower / faster
Ctrl+Enter	Approve & next (plays the next segment)
Ctrl+S	Save draft
Ctrl+Del	Reject
Alt+↑ / Alt+↓	Previous / next in queue
[]	Start –50 ms / +50 ms
{ }	End –50 ms / +50 ms
Alt+U/O/N/M/V	Toggle flags
Alt+E/S/X/L	Tag selected text as en / sw / mixed / Luo
R	Run the model on the segment
Ctrl+H	Copy hypothesis into the editor

The shortcuts are bound with a small injected JS snippet (`gr.Blocks(js=...)`), because Gradio has no native global hotkeys.

4.6 Corpus Studio: Datasets

- **Filter builder:** recordings (multi-select), domains, consent = training (locked), status = approved, exclude flags (default: unclear, non_verbatim, overlap), speaker roles (default: all), min/max duration.
- **Split:** “Use frozen test set v1” is locked. Dev % and seed are settable. A preview table shows each split's hours, segments, recordings and code-switch density, with a leakage check ✓.
- **Build snapshot** creates `kct-ds-v3 (frozen)`. **Export for Kaggle** produces a zip of the folder plus a README with the dataset hash.

4.7 Models & Evaluation

- **Registry table:** name, base model, dataset snapshot, dev WER, test WER, status (candidate/production/retired), and actions (Evaluate, Promote, Convert to CPU format).
- **Evaluation report:** metric cards (WER, CER, switch-point, hallucination/min, RTF), a breakdown table by domain × flag, and the worst 50 segments with audio, reference/hypothesis diff, and a “Send to review” action.

- **Compare:** two models on the same split, showing the per-metric delta, segments where A beats B and vice versa, and error categories (substitution en→sw and sw→en, deletions at switch points).

5 States and feedback

State	Treatment
Job queued/running	Progress bar with processed/total seconds and ETA; the page can be closed
Job failed	Red card with the error summary, a “Retry” action and the log link
No production model	Transcribe is disabled with the notice: “Register a model in Models & Evaluation”; Review still works without “Run model”
Empty review queue	“All segments reviewed. Build a dataset snapshot →”
Unsaved changes	A dot in the header, and a browser beforeunload warning
Offline aligner model missing	A setup checklist on first run (ffmpeg, aligner weights, VAD weights)

6 Accessibility

- Every action is keyboard-accessible, and the focus ring is a 2 px - - accent outline.
- Contrast is at least 4.5:1 for text in both themes (the tokens above meet this for body text on surface).
- Audio controls have labels; screen-reader text is set on the flag toggles.
- Playback speed runs from 0.5× to 1.5× to help with fast or accented speech.

7 Implementation notes

Changed in v1.1: see the change log.

The pages are plain HTML, CSS and JavaScript served by FastAPI, not Gradio. Gradio could not reliably capture F1–F4 (browsers use them for help, find and so on) or control playback speed, and the editor depends on both.

Need	Approach
Global hotkeys	One keydown listener that cancels the browser’s default for Esc and F1–F4
Speed	<code>HTMLAudioElement.playbackRate</code> (pitch preserved), kept when moving between chunks
Chunk audio	The server slices a 16 kHz WAV copy of the recording on request; the next chunk is preloaded
Background work	Two worker threads: full transcriptions, and look-ahead re-transcription after each save
Live updates	The editor checks every 4 s for chunks improved in the background
Waveform and boundary dragging (Corpus Studio, later)	A small canvas widget on the same page; no framework needed

Change log

Times are East Africa Time (EAT, UTC+3). Each entry names the affected section, quotes what it said before and summarises what it says now, and links to the previous text: on GitHub at the exact commit and lines, and as an archived PDF of the previous version.

Version	Date (EAT)	Commit	Summary
1.0	2026-09-24 00:36	053e09d	First LaTeX version (converted from the Markdown draft of 2026-09-23).
1.1	2026-10-02 00:34	this commit	Transcription app built: plain HTML/JS instead of Gradio; new Load audio and progress pages; chunk-by-chunk editor with Esc/F1–F4 keys and live adaptation; same keys in the reviewer.

v1.1 • Title block: Stack

Previous text: [v1.0 source, line 7](#) · [v1.0 PDF](#)

Before. “Gradio Blocks mounted inside FastAPI (v1). Custom React front end only if Gradio blocks a P0 requirement.”

Now. FastAPI serving a plain HTML/CSS/JavaScript page, no framework.

v1.1 • §4.1 Transcribe: New transcription

Previous text: [v1.0 source, lines 85–107](#) · [v1.0 PDF](#)

Before. Upload form with title, domain, model and consent checkbox; the user then lands on a jobs list with a progress bar.

Now. *Load audio* page (any format, preview, language, model, speaker labels, *Transcribe*) and a *Transcription in progress* page with stage, progress and time left that opens the editor by itself.

v1.1 • §4.2 Transcribe: Transcript editor

Previous text: [v1.0 source, lines 108–138](#) · [v1.0 PDF](#)

Before. One long transcript with a single player; Enter plays a segment, Ctrl+S saves, split/merge shortcuts, low-confidence underlines.

Now. Chunk-by-chunk editor: Esc play/pause, F1/F2 back/forward 3 s, F3/F4 slower/faster; moving saves and auto-plays; chunks improve while you work (rules, Whisper prompt, look-ahead re-transcription). Split/merge and low-confidence underlines are deferred.

v1.1 • §4.5 Segment reviewer: key map

Previous text: [v1.0 source, lines 223–238](#) · [v1.0 PDF](#)

Before. “Tab / Shift+Tab: play segment / replay last 3 s”.

Now. The same playback keys as the transcript editor: Esc, F1–F4; Ctrl+Enter also plays the next segment.

v1.1 • §7 Gradio feasibility notes → Implementation notes

Previous text: [v1.0 source, lines 303–319](#) · [v1.0 PDF](#)

Before. How each need would be met in Gradio (waveform, diff, span tagging, global hotkeys via injected JS, tabs).

Now. Why Gradio was dropped (F1–F4 and playback speed), and how each need is met in plain HTML/JS.